# CASE STUDY REPORT

## Cyclistic Bike-Share Analysis

Converting Casual Riders to Annual Members

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## Introduction

- **Project Overview:** A brief introduction to the Cyclistic company and the purpose of the study.
- **The Business Task:** Defining the primary goal—converting casual riders into annual members.

## Data Sources & Preparation

- **Data Description:** Details on the 12-month historical dataset (5.5 million rows).
- **Data Integrity (ROCCC):** Assessment of reliability, originality, and licensing.
- **Data Security:** Documentation on how PII (Personal Identifiable Information) was handled.

## Data Processing (PostgreSQL)

- Table Schema: Structure of the initial data import.

```sql
CREATE TABLE cyclistic_trips (
    ride_id VARCHAR(255) PRIMARY KEY,
    rideable_type VARCHAR(50),
    started_at TIMESTAMP,
    ended_at TIMESTAMP,
    start_station_name VARCHAR(255),
    start_station_id VARCHAR(100),
    end_station_name VARCHAR(255),
    end_station_id VARCHAR(100),
    start_lat FLOAT,
    start_lng FLOAT,
    end_lat FLOAT,
    end_lng FLOAT,
    member_casual VARCHAR(50)
);
```

- Cleaning Documentation: Handling NULL values and empty strings.

  *Removing trips < 1 minute and > 24 hours.

  *Data Transformation: Logic used to create ride_length and day_of_week.

| Date | Rows affected | Duration |
| --- | --- | --- |
| 20/2/2026 11:20:36 am | 2 | 2 secs 493 msec |

Copy | Copy to Query Editor

```sql
SELECT COUNT(*) AS count_outliers,member_casual
FROM cyclistic_trips
WHERE (ended_at - started_at) < INTERVAL '1 minute'
   OR (ended_at - started_at) > INTERVAL '24 hours'
GROUP BY member_casual;
```

```sql
CREATE TABLE cyclistic_final_analysis AS
SELECT
    ride_id,
    rideable_type,
    started_at,
    ended_at,
    (ended_at - started_at) AS ride_length,
    EXTRACT(DOW FROM started_at) AS day_of_week,
    member_casual,
    -- Labeling rows with missing stations so we can filter them easily later
    CASE
        WHEN start_station_name IS NULL OR end_station_name IS NULL THEN 'Missing Station'
        ELSE 'Complete'
    END AS data_quality_flag
FROM cyclistic_trips
WHERE (ended_at - started_at) > INTERVAL '1 minute'
  AND (ended_at - started_at) < INTERVAL '24 hours'
  AND member_casual IS NOT NULL;
```

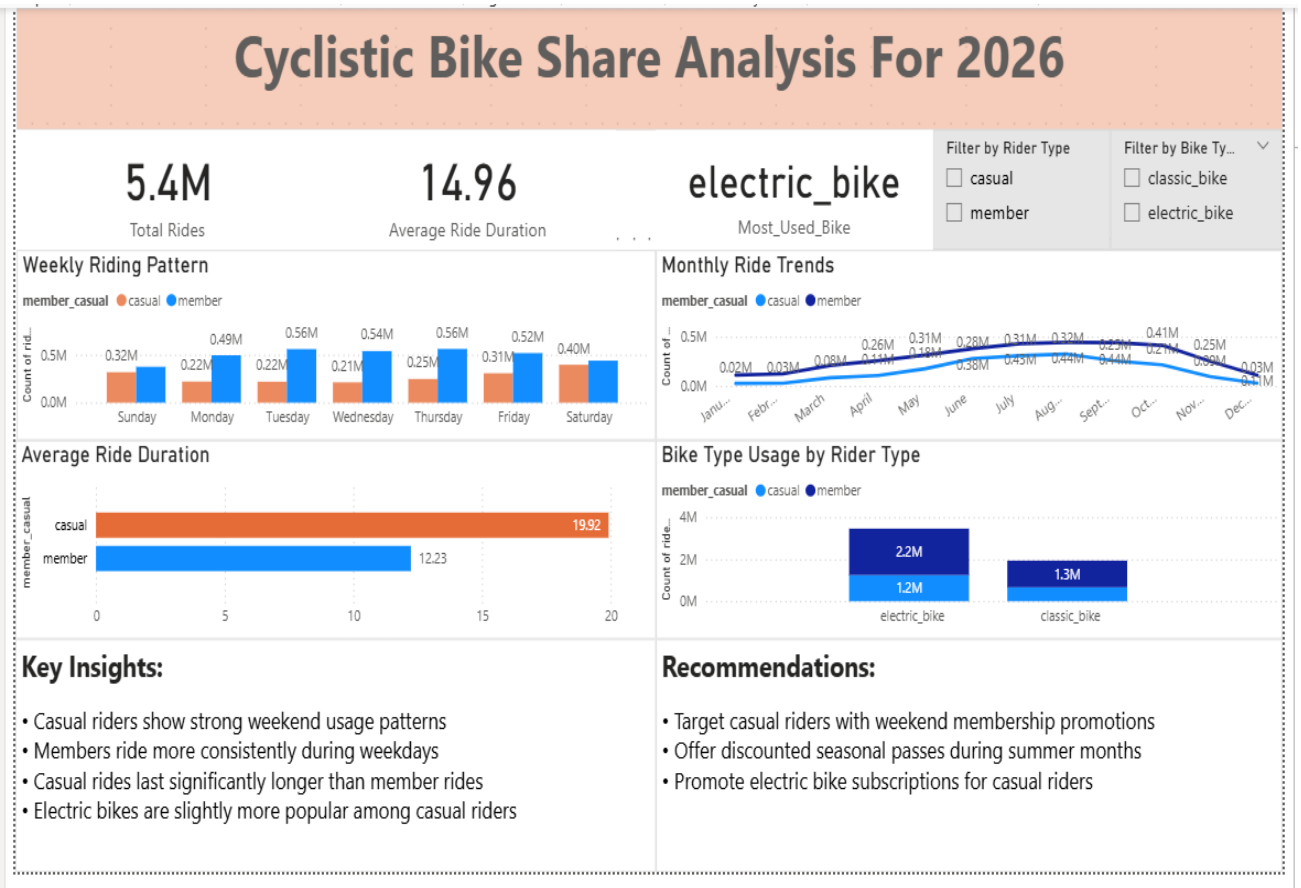
## Analysis & Insights

- **User Breakdown:** Total volume of Casuals (2M) vs. Members (3.5M).
- **Trip Duration Trends:** Analysis of the 19-minute vs. 12-minute average.

- **Temporal Patterns:** Weekly and seasonal ride distributions (Saturday peaks).
- **Geographic Analysis:** Most popular start/end stations for casual riders.

## Data Visualization (Power BI)

- **Dashboard Overview:** Screenshot of the primary interactive dashboard.




- **User Type Comparison:** Visualizing the 64% vs. 36% split.

**Key Insights:**

• Casual riders show strong weekend usage patterns
• Members ride more consistently during weekdays
• Casual rides last significantly longer than member rides
• Electric bikes are slightly more popular among casual riders

## Strategic Recommendations (The Act Phase)

The goal of this analysis was to identify how casual riders and annual members use Cyclistic bikes differently to create a conversion strategy. Based on the 5.5 million rows of data analyzed, the following three recommendations are proposed:

**Recommendation 1: The "Weekend Warrior" Annual Pass**

- **The Data Foundation:** Analysis revealed that casual ridership peaks on **Saturdays (Day 6)**, while members maintain a consistent weekday volume.
- **The Strategy:** Develop a specialized **"Weekend-Only Membership"** tier.

✓ **Detailed Action:**

*Many casual riders use Cyclistic for weekend leisure. A full-priced annual membership may feel like a "waste" to someone who doesn't commute via bike.

*A lower-cost, weekend-focused subscription lowers the barrier to entry and secures recurring revenue from the 2 million casual riders who currently pay per-ride.

**Recommendation 2: Membership Duration Perks (Extended Ride Limits)**

- **The Data Foundation:** The average ride length for casual riders is **19 minutes**, significantly higher than the **12-minute** average for members.
- **The Strategy:** Rebrand the annual membership as the **"Explorer's Pass,"** offering extended ride limits.

✓ **Detailed Action:**

*Currently, annual memberships are often optimized for short, 30-minute commuter bursts.

*By increasing the member time limit to 45 or 60 minutes, Cyclistic specifically targets the behavior of casual riders who prefer longer, recreational trips.

*Marketing this as "unlimited long-form riding" directly addresses the 19-minute usage habit of the casual segment.

**Recommendation 3: Targeted Station-Based Marketing Campaigns**

- **The Data Foundation:** Geographical analysis in Power BI identified that casual riders start and end their trips most frequently at stations near **tourist landmarks, parks, and the waterfront**, whereas members favor residential and business districts.
- **The Strategy:** Launch a **hyper-local digital marketing campaign** at top-performing casual stations.

✓ **Detailed Action:**

*Instead of a general city-wide ad campaign, use the PostgreSQL-derived "Top 10 Stations for Casual Riders" list.

| | start_station_name character varying (255) | total_rides bigint |
|---|---|---|
| 1 | DuSable Lake Shore Dr & Monroe St | 31292 |
| 2 | Navy Pier | 27590 |
| 3 | Streeter Dr & Grand Ave | 23542 |
| 4 | Michigan Ave & Oak St | 22401 |
| 5 | DuSable Lake Shore Dr & North Blvd | 19463 |
| 6 | Millennium Park | 19034 |
| 7 | Shedd Aquarium | 16885 |
| 8 | Theater on the Lake | 15711 |
| 9 | Dusable Harbor | 15599 |
| 10 | Michigan Ave & 8th St | 11184 |

**IMAGE: TOP 10 STATION LIST OF CASUAL RIDERS**

*Install physical QR codes at these docks that offer a **"Summer Conversion Discount."**
*When a casual rider docks their bike after a long Saturday ride at a high-traffic station, they receive a push notification or see an ad inviting them to convert their single-trip cost into a credit toward an annual membership.

## Conclusion

### Project Summary

This analysis of 5.5 million Cyclistic trip records successfully identified a clear behavioral divide between annual members and casual riders. By leveraging **PostgreSQL** for large-scale data processing and **Power BI** for trend visualization, we moved beyond raw numbers to uncover the "story" of the Cyclistic user.

**Key Takeaways**

- **The Commuter vs. The Explorer:** Annual members are "Utility" users, characterized by consistent, shorter trips (12-minute average) likely tied to work schedules. Casual riders are "Leisure" users, characterized by longer, spontaneous trips (19-minute average) that peak on weekends.
- **The Opportunity:** With over 2 million casual rides recorded, there is a massive opportunity to capture recurring revenue. The data shows that casual riders aren't just "one-time" users; they are frequent weekend explorers who have not yet seen the value in a traditional commuter-focused membership.

**Final Outlook**

By implementing the "Weekend Warrior" membership and targeting high-traffic tourist stations, Cyclistic can transition from a service that casual riders use *occasionally* to a lifestyle brand they subscribe to annually. These data-driven strategies provide a clear roadmap for the Marketing Department to increase profit margins and ensure long-term company sustainability.

**Final Summary (Author's Note)**

Completing this capstone project allowed me to apply the full scope of the data analytics lifecycle—from initial problem definition to the delivery of actionable business insights.

Using **PostgreSQL** allowed me to manage a "Big Data" environment efficiently, while **Power BI** enabled me to transform complex SQL aggregations into a narrative that stakeholders can act upon.

As a Computer Science and Engineering student at **Anna University (BIT Campus)**, this project has strengthened my ability to bridge the gap between technical data engineering and strategic business analysis.

# Appendix

SQL CODE SNIPPETS:

## A. Data Transformation (The "Process" Phase)

The following query shows the logic used to create the calculated columns for duration and time-based analysis.

```
/* Calculating ride length in minutes and extracting day of week/month from the raw timestamps */

 SELECT

   ride_id,

   member_casual,

   -- Calculate duration in minutes by extracting epoch seconds

   EXTRACT(EPOCH FROM (ended_at - started_at)) / 60 AS ride_length_m,

   -- Extract Day of Week (0 = Sunday, 6 = Saturday)

   EXTRACT(DOW FROM started_at) AS day_of_week,

   -- Extract Month for seasonality analysis

   EXTRACT(MONTH FROM started_at) AS month_num

FROM cyclistic_trips

WHERE

   (ended_at - started_at) > INTERVAL '1 minute' -- Filtering false starts

   AND (ended_at - started_at) < INTERVAL '24 hours'; -- Removing outliers
```

## Data Aggregation (The "Analyze" Phase)

This query was used to find the **Top 10 Stations** for casual riders, which supported Recommendation #3.

```
/* Identifying high-traffic stations for casual riders to target  for marketing
campaigns.*/

 SELECT

   start_station_name,

   COUNT(*) AS total_rides

FROM cyclistic_raw_data

WHERE

   member_casual = 'casual'

   AND start_station_name IS NOT NULL

GROUP BY

   start_station_name

ORDER BY

   total_rides DESC

LIMIT 10;
```